LAB WRITEUP 03 - INTRO TO LOG ANALYSIS (TRYHACKME)

Room	Intro to Log Analysis
Platform	TryHackMe - `https://tryhackme.com/room/introtologanalysis`
Path	SOC Level 1 / SOC Level 2 - Log Analysis module (final room)
Difficulty	Easy
Time to complete	About 60-90 minutes
Tasks	9

Objective

Turn raw log files into answers. The room covers investigation theory, detection engineering, the trade-off between manual and automated analysis, and four analyst tools: the Linux command line, regular expressions, CyberChef, and the detection-rule languages YARA and Sigma.

Concepts covered

- Timelines and the "super timeline" approach
- Threat-intelligence indicators and common attack signatures
- Manual versus automated analysis
- `grep`, `awk`, `cut`, `sort`, `uniq`, `wc` on an Apache access log
- Regular expressions for filtering logs
- CyberChef for decoding and extracting from logs
- Sigma (YAML) and YARA (rule) detection languages

Task 1 - Introduction / Task 2 - What is log analysis?

No answers required. Log analysis is defined as the practice of reading logs not to admire them but to answer a question - usually "was this malicious, and what happened next?".

Task 3 - Investigation Theory

Question	Answer	How
Term for a consolidated chronological view of logged events from diverse sources	Super Timeline (also called a consolidated timeline)	Each individual source has its own timeline; the value for an investigation comes from merging them into one timeline so an event on a workstation can be lined up against the firewall, proxy and DNS logs at the same second.
Which threat-intelligence indicator type are `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe`?	File hashes	Both strings are 32 hexadecimal characters, which is the fixed length of an MD5 digest. A hash identifies a file's content, so the same value can be searched in VirusTotal or a threat-intel feed and matched against other hosts.

Other indicator types worth remembering: IP addresses, domains, URLs, email addresses, registry keys and file names.

Task 4 - Detection Engineering

Question	Answer	How
Default path for HTTP request logs on an Nginx server	`/var/log/nginx/access.log`	Nginx writes access entries to `access.log` and error entries to `error.log`; both live under `/var/log/nginx/`.
Attack inferred by a log entry containing `%2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron`	Path traversal	The value is URL-encoded; `%2E` is `.` and `%2F` is `/`, which decodes to `../../proc/self/environ`. Walking up the directory tree with `../` to reach a file outside the web root is the definition of path traversal (and reaching `/proc/self/environ` specifically aims at leaking environment variables such as credentials).

Detection-engineering habit: when you find a pattern like this in an alert, do not stop at "one bad request". Decode it, name the attack, then search the log for every other occurrence of the same decoded string. Attackers almost never try an exploit exactly once.

Task 5 - Automated vs manual analysis

Question	Answer	How
A log file is processed by a tool which returns an output. What form of analysis is this?	Automated	A tool (SIEM rule, script, parser) does the matching at machine speed. Great coverage and consistency; useless if nobody wrote a rule for the behaviour.
An analyst opens a log file and searches for events. What form of analysis is this?	Manual	Human-driven reading and searching. Slow and hard to repeat, but it is how a new or unusual behaviour is found for the first time - the finding then gets turned into an automated rule.

The rule of thumb: automate the known, investigate manually the unknown.

Task 6 - Log Analysis Tools: Command Line

Working on `apache.log` with `grep`, `awk`, `cut`, `sort`, `uniq` and `wc`.

Question	Answer	Command used			
Flag returned in one of the unique URLs	c701d43cc5a3acb9b5b04db7f1be94f6	`cut -d ' ' -f 7 apache.log` returns field 7, the requested URL, from every line; one request is made to a URL containing the flag.			
Total number of HTTP 200 responses logged	52	`awk '$9 == 200' apache.log \`	`wc -l - field 9 holds the status code, and wc -l` counts the matching lines.		

Question	Answer	Command used			
IP address that generated the most traffic	145.76.33.201	`cut -d'' -f 1 apache.log \	sort -n \	uniq -c \	sort -nr` - take field 1 (client IP), sort, count duplicates and sort by count. The highest count (8 requests) belongs to this address.
Complete timestamp of the entry where `110.122.65.76` accessed `/login.php`	`31/Jul/2023:12:34:40 +0000`	`grep -E '110.122.65.76.*/login.php' apache.log` matches the one line containing both values; the timestamp is field 4 of that line.			

Technique note: `grep -E 'a.*b'` is how an "AND" search is expressed in grep - both patterns must appear, in that order, on the same line. With hundreds of thousands of lines in a real log, this single command is often what identifies the attacker's session.

Task 7 - Log Analysis Tools: Regular Expressions

Question	Answer	How
How to modify `grep -E 'post=1[0-9]'` to match blog posts with an ID between 22 and 26	`grep -E 'post=2[2-6]' apache.log`	`[2-6]` is a character class matching any single digit from 2 to 6, so the pattern matches `post=22` through `post=26` and nothing else.
Name of the Logstash filter plugin used to parse unstructured log data	Grok	Grok patterns use named regular expressions such as `%{IP}` and `%{TIMESTAMP_ISO8601}` so that unstructured lines can be turned into structured fields during ingestion.

Task 8 - Log Analysis Tools: CyberChef

CyberChef is used as a recipe pipeline: a regex operation to extract, then further operations to filter or decode.

Question	Answer	How	
Full IP address beginning with `212` in `access.log`	212.14.17.145	Apply the "Regular expression" operation with the IPv4 pattern, then narrow it by changing the first octet, i.e. `\b212\.([0-9]{1,3}\.){2}[0-9]{1,3}\b`, and set the output to list all matches.	
Decoded value of the Base64-encoded request	THM{CYBERCHEF_WIZARD}	Extract candidate strings with `(POST\`	GET)/.*=, then run the `Base64 string` VEhNe0NZQkVSQ0hFRl9XSVpBUkR9` through the "From Base64" operation.

Question	Answer	How	
MAC address extracted from `encodedflag.txt`	08-2E-9A-4B-7F-61	Base64-decode the file, then apply the built-in MAC-address regular expression. The surrounding decoy values contain an extra hexadecimal group, so only the strict pattern returns the real one.	

Why this belongs in a SOC: base64 is the simplest obfuscation attackers use, so an analyst must be able to decode it instantly rather than treating the string as unreadable noise.

Task 9 - Log Analysis Tools: YARA and Sigma

Question	Answer	How
What language does Sigma use?	YAML	Sigma rules are written in YAML so they stay readable and portable.
Keyword that denotes the "title" of a Sigma rule	`title`	The first field of a Sigma rule, followed by `id`, `status`, `description`, `logsource`, `detection` and `level`.
Keyword that denotes the "name" of a rule in YARA	`rule`	A YARA rule is declared as `rule RuleName { ... }`, with `meta`, `strings` and `condition` sections.

Sigma vs YARA: Sigma describes behaviour in logs and is converted into queries for whichever SIEM is in use, which makes detections portable. YARA describes patterns in files and memory and is used for scanning and malware classification.

Key takeaways

1. Build a super timeline - a single source rarely tells the whole story.
2. MD5/SHA hashes are indicators, and they identify content rather than location, which is what makes them searchable across every host.
3. Encoded strings in a URL are never an accident - decode first, then name the attack.
4. A handful of command-line tools (`grep`, `awk`, `cut`, `sort`, `uniq`, `wc`) can answer who, how many, and from where in seconds.
5. CyberChef turns obfuscated log content back into readable evidence.
6. Sigma (YAML) is for log-based detections; YARA is for file/memory patterns.